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 Advanced Biostatistical Analysis
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Final Project Writeup – Heart Disease Prediction Dataset

Background and Motivation:

As public health concerns rise, heart disease seems to be one of the top contenders to these concerns. It is known to be one of the leading causes of global mortality. Early prediction and diagnosis of heart disease is crucial, to reduce its risk and prevalence. It is known that there are multiple factors that influence heart disease, which is why there are many tests and tools used to perform diagnostics. It is also important to know how clinical indicators can predict heart disease.

Based on the heart disease risks dataset, the scientific question that I am addressing is: **Does adding additional categorical risk factors improve the prediction of heart disease compared to using chest pain type alone?** My motivation behind this analysis was to evaluate whether prediction improves when including and adding multiple categorical indicators. I wanted to see the difference it would make compared to being reliant on one singular indicator, which in this case would be considered the chest pain type. A lot of analyses in the past have shown strong association with the chest pain type predictor, however, many studies have shown that there is a variability to the factors that influence heart disease. The information that I am seeking in my analysis is to determine if a multivariable logistic regression model outperforms a single-predictor model that is primarily based on chest pain type and no other predictors. I want to evaluate the performance of each logistic regression model and see if multiple variables allow for better accuracy in the data that is used to assist in heart disease detection.

Description of Dataset and Variables:

The data source is originally sourced from the Cleveland Heart Disease dataset from the UCI Machine Learning Repository. The main purpose of this data source and dataset is to predict whether a patient has heart disease or not, based on specific characteristics they may have. This data source has a sample size of 920 patient records and has a total of 14 key variables which form a subset of variables out of 76 attributes. Although there are 16 columns in the dataset, only 14 of these columns represent the key variables. The following will show the supporting documentation that describes the variables:

- id = unique id for each patient
- age = age of the patient in years
- origin = place of study
- sex =
 - 1 = male
 - 0 = female
- cp = chest pain type (typical angina, atypical angina, non-anginal, asymptomatic)
 - 1 - 4
- trestbps = resting blood pressure (in mmHg on admission to the hospital)
- chol = serum cholesterol in mg/dl
- fbs = if fasting blood sugar > 120 mg/dl
 - 1 = true

- 0 = false
- restecg = resting electrocardiographic results (normal, stt abnormality, lv hypertrophy)
 - 0 – 2
- thalch = maximum heart rate achieved
- exang = exercise-induced angina (true or false)
 - 1 = true
 - 0 = false
- oldpeak = ST depression induced by exercise relative to rest
- slope = the slope of the peak exercise ST segment
- ca = number of major vessels (0-3) colored by fluoroscopy
- thal = normal, fixed defect, reversible defect
 - 3 = normal
 - 6 = fixed defect
 - 7 = reversible defect
- num = the predicted attribute (0 – 4)
 - 0 = no disease
 - 1 – 4 = disease present (range starts from 1 = less severe to 4 = most severe)

As for this specific study, the inclusion criteria consist of adults who had medical examinations that were prevalent to the functions of their cardiac health, at the Cleveland Clinic Foundation. The exclusion criteria consisted of patients who didn't complete all the diagnostic medical examination or if they had missing test results that were crucial to the data prediction model. This is a cross-sectional study design, specifically an observational one. This is because each patient has their own attributes, and this data is collected from a specific and defined population. All the data is then used to determine the prevalence of heart disease in each patient. The structure of this study is what makes it a cross-sectional study design.

The variables that I will be using are only the categorical variables from all the variables shown above, in the original dataset:

- sex =
 - 1 = male
 - 0 = female
- cp = chest pain type (typical angina, atypical angina, non-anginal, asymptomatic)
 - 1 – 4
- fbs = if fasting blood sugar > 120 mg/dl
 - 1 = true
 - 0 = false
- exang = exercise-induced angina (true or false)
 - 1 = true
 - 0 = false
- restecg = resting electrocardiographic results (normal, stt abnormality, lv hypertrophy)
 - 0 – 2
- slope = the slope of the peak exercise ST segment
- ca = number of major vessels (0-3) colored by fluoroscopy
- thal = normal, fixed defect, reversible defect
 - 3 = normal
 - 6 = fixed defect
 - 7 = reversible defect

The outcome variable is num. When num = 0, there is no heart disease. However, if it ranges from 1-4, that means there is heart disease that ranges from 1 being less severe to 4 being the most severe. This outcome variable was converted into a binary variable to show the status of the disease. Within scientific context, this means each variable represents a common diagnostic marker that is used in cardiology to classify ischemia, any structural abnormalities, and metabolic risk. As seen above, each variable is described in its scientific classifications.

Description of Methods:

The statistical methods and framework that I will use is based on a logistic regression approach. I made my analysis based off this because the outcome variable was binary (no disease vs. disease). I think a logistic regression approach is appropriate for the data and research question because it produces odds ratios, deals with categorical predictors, and is a good base for clinical classifications. It also accounts for multivariable regression which holds multiple predictors.

For my statistical analysis plan, I will start off by importing the dataset and cleaning the data. The cleaning requires converting some predictors into categorical variables, while also making the outcome variables into a binary variable. The initial statistical method to perform a hypothesis test will be done using chi-squared tests. This will allow me to check for unadjusted associations between the categorical predictors and the outcome variable. After this, I will develop 4 logistic regression models since the outcome variable is binary. Logistic regression creates graphical and numerical explanations, like odds ratio, predicted probabilities, and confidence intervals. This creates an analysis that shows the association between the predictor and outcome variable for heart disease. The first model will focus just on chest pain type by itself. As the models change, I will keep adding more indicators, as shown in what each model is going to demonstrate below:

- Model 1: only cp
 - Equation $\rightarrow \text{logit}(P(\text{Disease})) = B_0 + B_1(\text{cp})$
- Model 2: cp + sex $\rightarrow$ determines the demographic differences contribution to predictive value
- Model 3: cp + sex + fbs $\rightarrow$ metabolic risk impact on prediction
- Model 4: cp + sex + fbs + exang + restecg + slope + ca + thal $\rightarrow$ overall predictive impact
 - Equation $\rightarrow \text{logit}(P(\text{Disease})) = B_0 + B_1(\text{cp}) + B_2(\text{sex}) + B_3(\text{fbs}) + B_4(\text{exang}) + B_5(\text{restecg}) + B_6(\text{slope}) + B_7(\text{ca}) + B_8(\text{thal})$

Since chest pain type had the strongest association and is a primary predictor in past studies, that is the variable I chose to focus on in Model 1. Sex is added in Model 2 because it is clinically significant to heart disease. Model 3 adds fbs because it shows the metabolic risk impact on the predictive accuracy. Model 4 includes all categorical variables, however, visually the graph only shows cp, sex, and thal. Thal has the highest odds ratio out of all the categorical predictors. Although mathematically all categorical variables are account for in Model 4, it cannot all be graphed onto 1 plot because it would be unreadable. So, the model focuses on the 3 main variables. I will create diagnostic plots to check for different patterns that can potentially be indicators for a poor fit. These diagnostic plots include Pearson residuals in comparison to fitted values. This will show us if the models are appropriate for the data and their overall

performance. I will also compare the models using AIC values and likelihood ratio tests. The lower the AIC is, the better the fit is. This will show the impact that adding indicators has on the accuracy of predictivity. These statistical methods, exploratory data analysis, and different summaries of the data, will collectively add to the analysis of the data and show the contribution that each categorical variable has on the outcome variable. It will also prove if a multivariable regression model is more impactful than a single linear regression model. Some assumptions related to logistic regression are that the outcome must be binary and to have a stable analysis outcome, the sample size must be big enough. In addition, the predictors don't have exact multicollinearity. Another assumption is each patient record is equivalent to a specific patient.

Results:

Figure 1: chi-squared tests

```
> #chi-square tests:
>
> #disease vs. cp
> heart_cp <- heart %>%
+   filter(!is.na(disease), !is.na(cp))
>
> tab_cp <- table(heart_cp$disease, heart_cp$cp)
> tab_cp
```

	asymptomatic	atypical	angina	non-anginal	typical	angina
No disease	104		150	131		26
Disease	392		24	73		20

```
> chisq.test(tab_cp)
```

Pearson's Chi-squared test

data: tab_cp
X-squared = 268.35, df = 3, p-value < 2.2e-16

```
>
> #disease vs. sex
> heart_sex <- heart %>% drop_na(disease, sex)
> tab_sex <- table(heart_sex$disease, heart_sex$sex)
> print(tab_sex)
```

	Female	Male
No disease	144	267
Disease	50	459

```
> chisq.test(tab_sex)
```

Pearson's Chi-squared test with Yates' continuity correction

data: tab_sex
X-squared = 85.361, df = 1, p-value < 2.2e-16

```
>
> #disease vs. fbs
> heart_fbs <- heart %>% drop_na(disease, fbs)
> tab_fbs <- table(heart_fbs$disease, heart_fbs$fbs)
> print(tab_fbs)
```

	FALSE	TRUE
No disease	353	44
Disease	339	94

```
> chisq.test(tab_fbs)
```

Pearson's Chi-squared test with Yates' continuity correction

data: tab_fbs
X-squared = 16.112, df = 1, p-value = 5.972e-05

```
> #disease vs. exang
> heart_exang <- heart %>% drop_na(disease, exang)
> tab_exang <- table(heart_exang$disease, heart_exang$exang)
> print(tab_exang)
```

	FALSE	TRUE
No disease	336	55
Disease	192	282

```
> chisq.test(tab_exang)
```

Pearson's Chi-squared test with Yates' continuity correction

data: tab_exang
X-squared = 184.02, df = 1, p-value < 2.2e-16

```
>
> #disease vs. restecg
> heart_restecg <- heart %>% drop_na(disease, restecg)
> tab_restecg <- table(heart_restecg$disease, heart_restecg$restecg)
> print(tab_restecg)
```

	lv hypertrophy	normal	st-t abnormality
No disease	82	268	61
Disease	106	283	118

```
> chisq.test(tab_restecg)
```

Pearson's Chi-squared test

data: tab_restecg
X-squared = 11.712, df = 2, p-value = 0.002863

```
>
> #disease vs. slope
> heart_slope <- heart %>% drop_na(disease, slope)
> tab_slope <- table(heart_slope$disease, heart_slope$slope)
> print(tab_slope)
```

	downsloping	flat	upsloping
No disease	14	79	125
Disease	49	266	78

```
> chisq.test(tab_slope)
```

Pearson's Chi-squared test

data: tab_slope
X-squared = 88.852, df = 2, p-value < 2.2e-16

```

>
> #disease vs. ca
> heart_ca <- heart %>% drop_na(disease, ca)
> tab_ca <- table(heart_ca$disease, heart_ca$ca)
> print(tab_ca)

      0  1  2  3
No disease 133 21 8 3
Disease 48 46 33 17
> chisq.test(tab_ca)

Pearson's Chi-squared test

data:  tab_ca
X-squared = 73.2, df = 3, p-value = 8.806e-16

>
> #disease vs. thal
> heart_thal <- heart %>% drop_na(disease, thal)
> tab_thal <- table(heart_thal$disease, heart_thal$thal)
> print(tab_thal)

      fixed defect normal reversable defect
No disease      11    138              38
Disease        35     58             154
> chisq.test(tab_thal)

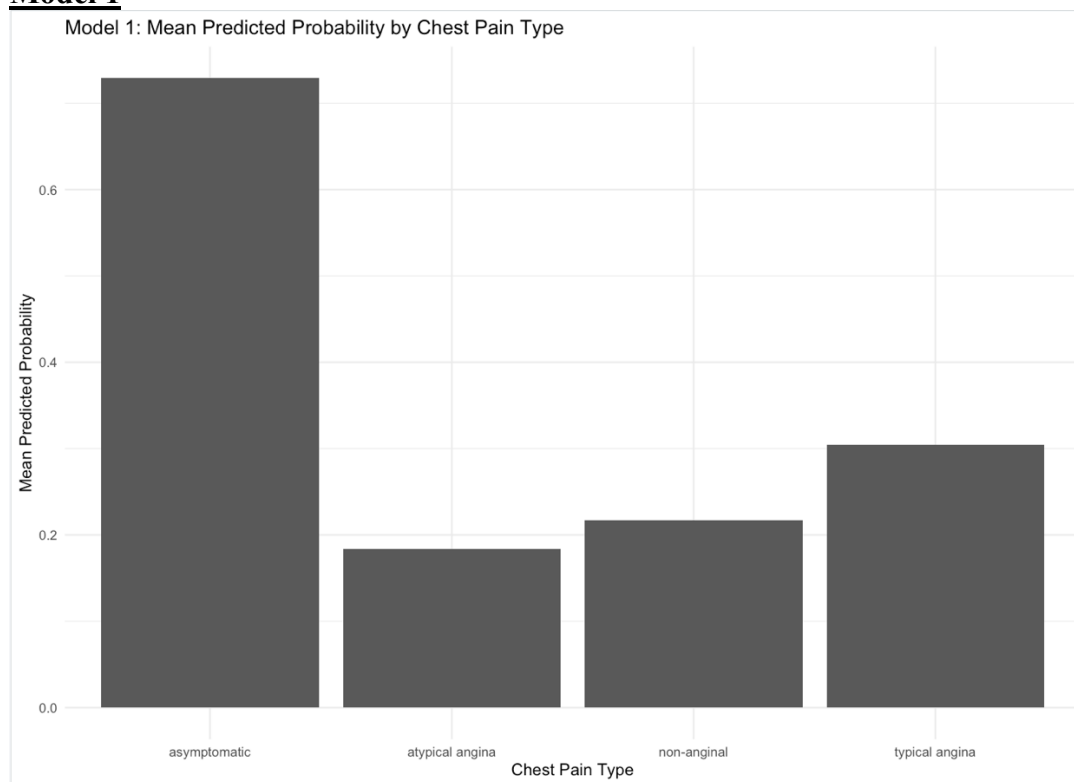
Pearson's Chi-squared test

data:  tab_thal
X-squared = 109.05, df = 2, p-value < 2.2e-16

```

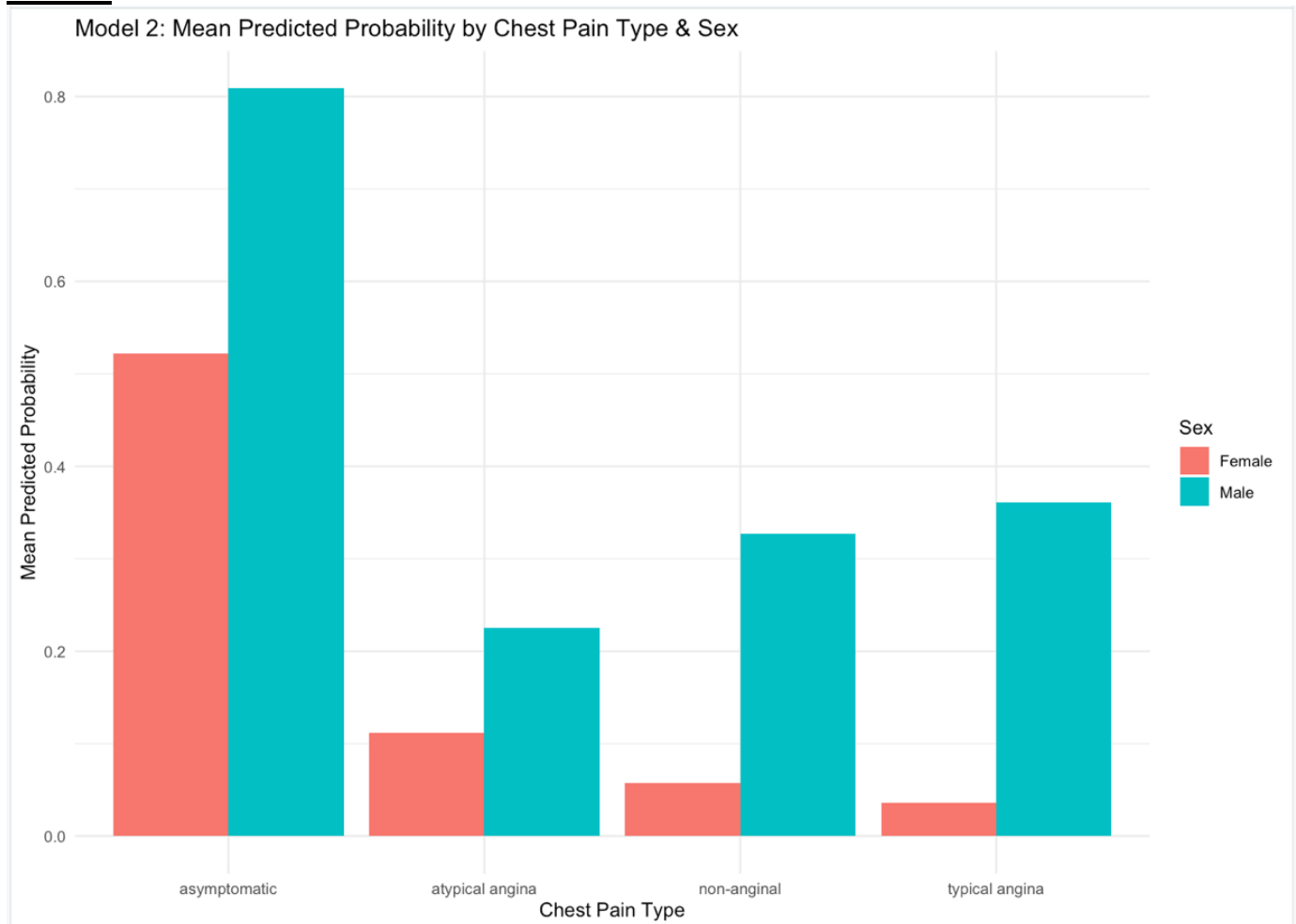
Figure 1 Explanation → After conducting a chi-square test for each categorical variable, each contingency table displays a comparison between heart disease and no heart disease to the categorical variables. Besides restecg, every other variable showed large chi-square values. All the categorical variables had p-values less than 0.001 which demonstrated that heart disease was not evenly spread out across each predictor. Chest pain type, sex, exercise-induced angina, slope, ca, and thal, were the variables that displayed a strong unadjusted association with the outcome variable of having heart disease. Having a strong chi-squared value and test proves that a multivariable regression model would be a lot more impactful in showing the differences in risk for heart disease amongst all the categorical predictors.

Model 1

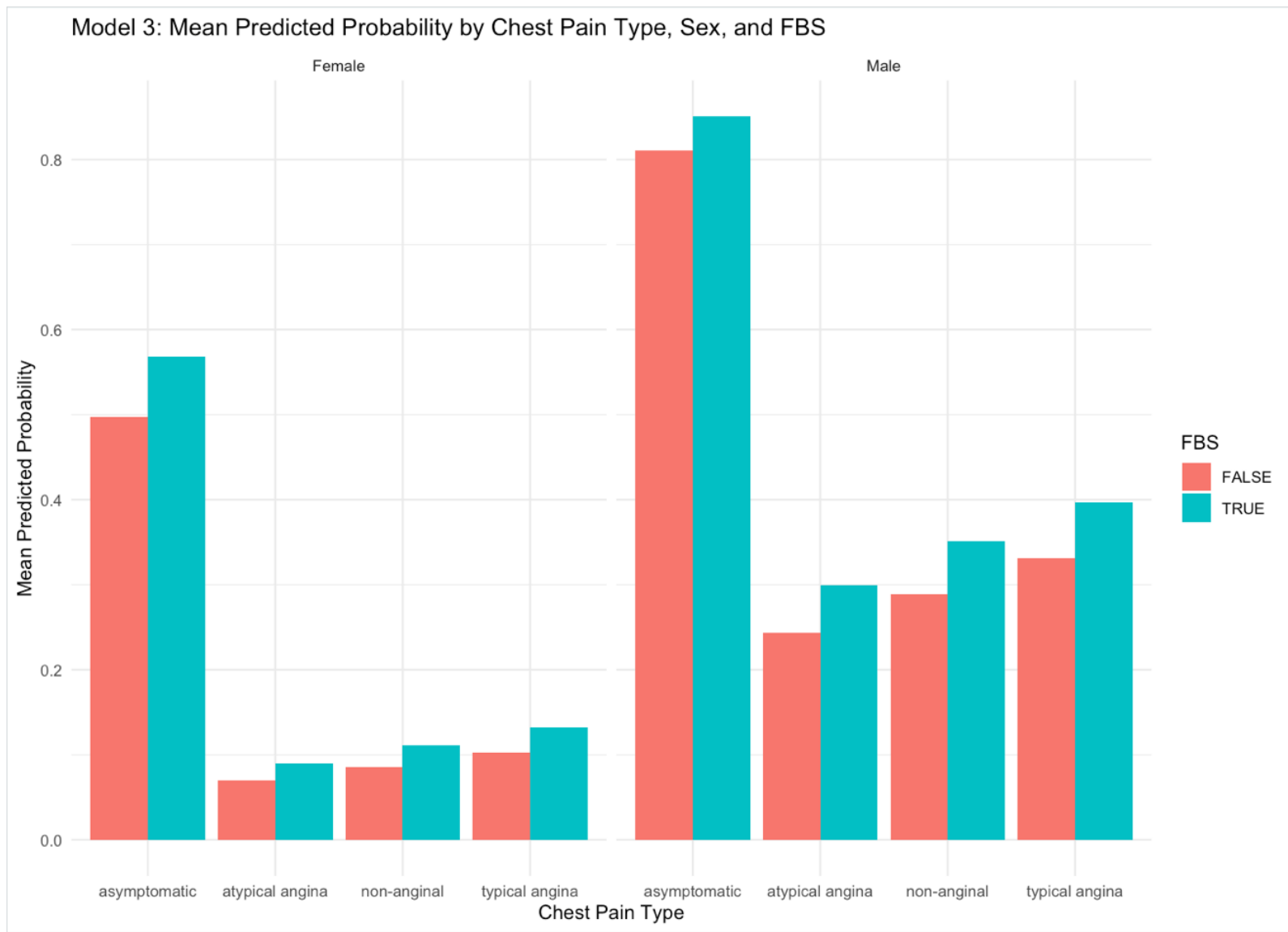


Model 1 Explanation → This bar plot displays the mean probability of disease for each category in chest pain. The highest predicted probability is found in asymptomatic chest pain. This graph matches the odds ratio because that shows low ORs for all the anginal types of chest pain. This model shows that patients who are asymptomatic have the highest risk for heart disease. This shows a limitation because when only using chest pain as a predictor to assess heart disease, you get a very isolated outcome which isn't as strong, compared to having multiple predictors.

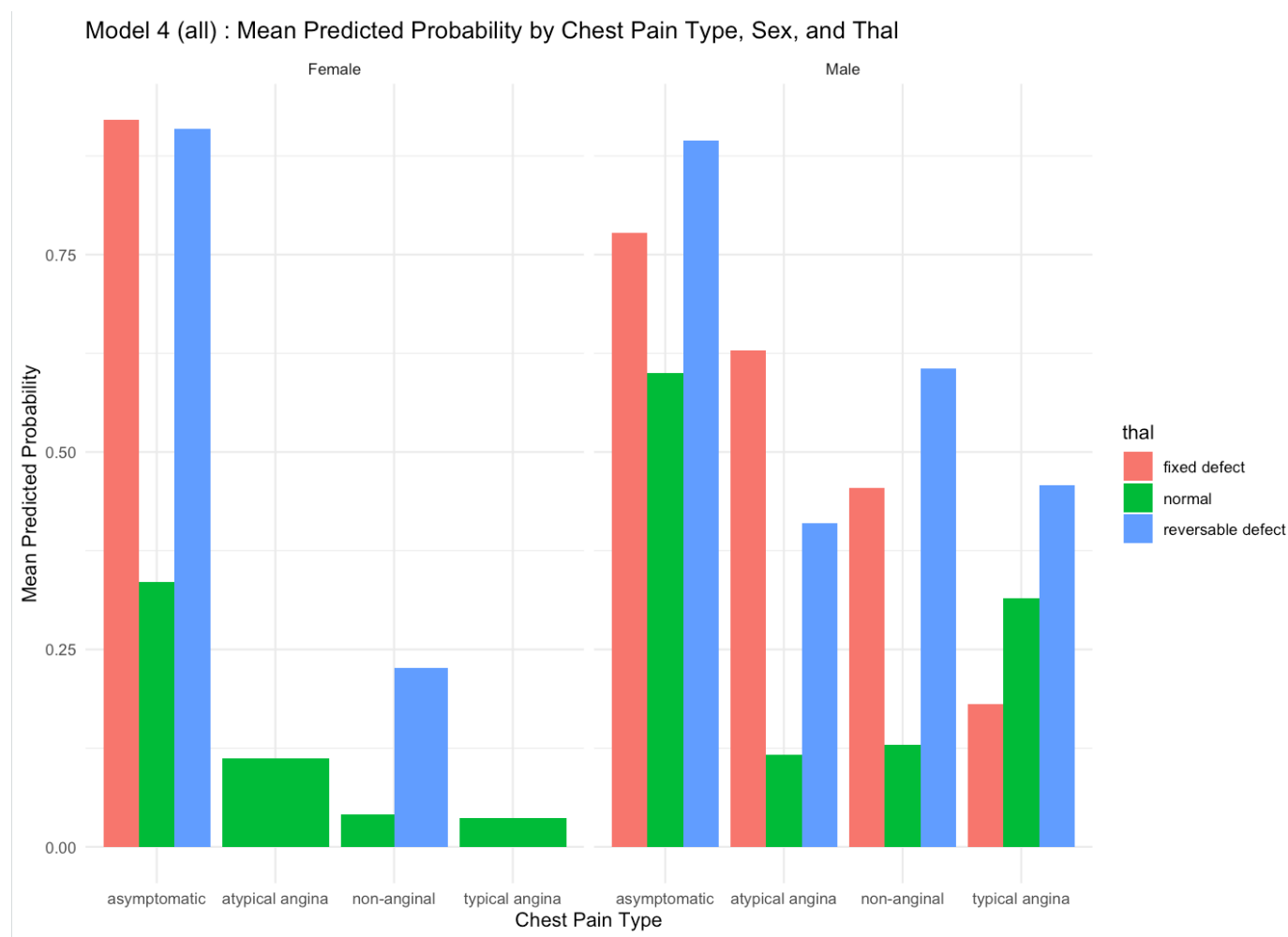
Model 2



Model 2 Explanation → This bar plot compares the predicted disease probability for males vs. females and their chest pain type. It is visible in the graph that males have a higher predicted probability of disease. Females have a lower probability even when the chest pain type is of a higher risk. Adding the sex improves the accuracy of predictability because it is a strong independent predictor. This increases the odds of disease.

Model 3

Model 3 Explanation → This bar plot shows the predicted risk for a combination of chest pain type and fasting blood sugar amongst females and males. Adding the fasting blood sugar (fbs) doesn't make the biggest difference in the risk patterns that were created by cp and sex. This model shows that fbs is not as strong of a predictor because it doesn't change much.

Model 4

Model 4 Explanation → For model 4, it only visually focuses on chest pain type, sex, and thal because those are the strongest categorical predictors. However, mathematically, the code for this model accounts for all categorical variables. It just isn't shown in the graph because it can't all be graphed, since there are too many combinations, thus making the graph unreadable. So, this graph demonstrates the predicted probabilities grouped by chest pain type and sex, while being colored in by the thalassemia categories. The outcome of thal created a drastic change which separated the risk groups. The highest predicted probability was seen in the reversible defects. Low risk is seen in normal thal across all groups. The overall outcome showed that thal is one of the strongest predictors which proves how multivariable regression models can analyze more deeply and show more thorough patterns.

Figure 2: odds ratio & 95% CIs

```

> tidy(mod1, exponentiate = TRUE, conf.int = TRUE)
# A tibble: 4 × 7
  term                estimate std.error statistic p.value conf.low conf.high
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)          2.69      0.188      5.28 1.28e- 7    1.88      3.93
2 cpatypical angina    0.0836    0.414    -6.00 2.01e- 9    0.0352    0.181
3 cpnon-anginal        0.103      0.326    -6.98 2.90e-12    0.0531    0.191
4 cptypical angina     0.163      0.490    -3.71 2.11e- 4    0.0586    0.411
> tidy(mod2, exponentiate = TRUE, conf.int = TRUE)
# A tibble: 14 × 7
  term                estimate std.error statistic p.value conf.low conf.high
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)          0.391      1.08    -0.869 0.385    0.0457    3.26
2 cpatypical angina     0.360      0.536    -1.90 0.0568    0.121     1.00
3 cpnon-anginal         0.158      0.483    -3.82 0.000131 0.0590    0.396
4 cptypical angina     0.167      0.608    -2.95 0.00319 0.0480    0.532
5 sexMale               3.53      0.475      2.65 0.00797    1.42     9.25
6 fbsTRUE                0.614      0.564    -0.863 0.388    0.197     1.82
7 exangTRUE              2.51      0.414      2.22 0.0264    1.11     5.68
8 restecgnormal         0.569      0.365    -1.55 0.122    0.275     1.16
9 restecgst-t abnormality 4.18      1.78      0.801 0.423    0.140    175.
10 slopeflat            1.35      0.701      0.431 0.667    0.338     5.35
11 slopeupsloping       0.281      0.720    -1.76 0.0783    0.0666    1.14
12 ca                   3.77      0.253      5.24 0.000000159 2.36     6.39
13 thalnormal            0.757      0.747    -0.373 0.709    0.171     3.28
14 thalreversible defect 3.78      0.726      1.83 0.0667    0.895    15.9
> tidy(mod3, exponentiate = TRUE, conf.int = TRUE)
# A tibble: 6 × 7
  term                estimate std.error statistic p.value conf.low conf.high
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)          0.990      0.282    -0.0368 9.71e- 1    0.566     1.72
2 cpatypical angina     0.0751    0.433    -5.98 2.19e- 9    0.0305    0.169
3 cpnon-anginal         0.0950    0.350    -6.73 1.68e-11 0.0466    0.185
4 cptypical angina     0.116      0.515    -4.19 2.82e- 5    0.0398    0.307
5 sexMale               4.33      0.322      4.54 5.54e- 6    2.33     8.30
6 fbsTRUE                1.33      0.396      0.717 4.73e- 1    0.610     2.90
> tidy(mod4, exponentiate = TRUE, conf.int = TRUE)
# A tibble: 14 × 7
  term                estimate std.error statistic p.value conf.low conf.high
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)          0.391      1.08    -0.869 0.385    0.0457    3.26
2 cpatypical angina     0.360      0.536    -1.90 0.0568    0.121     1.00
3 cpnon-anginal         0.158      0.483    -3.82 0.000131 0.0590    0.396
4 cptypical angina     0.167      0.608    -2.95 0.00319 0.0480    0.532
5 sexMale               3.53      0.475      2.65 0.00797    1.42     9.25
6 fbsTRUE                0.614      0.564    -0.863 0.388    0.197     1.82
7 exangTRUE              2.51      0.414      2.22 0.0264    1.11     5.68
8 restecgnormal         0.569      0.365    -1.55 0.122    0.275     1.16
9 restecgst-t abnormality 4.18      1.78      0.801 0.423    0.140    175.
10 slopeflat            1.35      0.701      0.431 0.667    0.338     5.35
11 slopeupsloping       0.281      0.720    -1.76 0.0783    0.0666    1.14
12 ca                   3.77      0.253      5.24 0.000000159 2.36     6.39
13 thalnormal            0.757      0.747    -0.373 0.709    0.171     3.28
14 thalreversible defect 3.78      0.726      1.83 0.0667    0.895    15.9
> |

```

Figure 2 Explanation → This figure demonstrates the odds ratio for each model. In Model 1, anginal chest pain types have ORs that ranges from 0.1 to 0.2. In Model 2, the odds ratio for male sex ranges from 3.5-4.0 which means that it has a strong effect. In Model 3, the odds ratio of fbs is 1.3, which means it is weaker. This is also because the confidence interval includes 1. In Model 4, the confidence intervals of thal, exang, ca, and sex are narrower, and their odds ratios are larger. The odds ratio shows how each predictor is associated with the disease risk and how

strong of an impact each predictor has. This shows how chest pain type by itself is not as efficient as a model that accounts for all the strongest predictors.

Figure 3: comparing AIC

```
# A tibble: 4 x 2
  Model                                     AIC
  <chr>                                <dbl>
1 Model 1 : Mean Predicted Probability by Chest Pain Type 338.
2 Model 2 : Mean Predicted Probability by Chest Pain Type & Sex 232.
3 Model 3: Mean Predicted Probability by Chest Pain Type, Sex, and FBS 319.
4 Model 4 (all) : Mean Predicted Probability by Chest Pain Type, Sex, and Thal 232.
> |
```

Figure 3 Explanation → Model 1 has the highest AIC (338), which means it has the worst fit. Model 4 and Model 2 have the lowest AIC (232), which means they have the best fit. The lower the AIC, the better the fit is. This shows how including more than one predictor improves the predictive accuracy.

Figure 4: ratio test

```
> #ratio test
> anova(mod1, mod2, test = "Chisq")
Analysis of Deviance Table

Model 1: disease ~ cp
Model 2: disease ~ cp + sex + fbs + exang + restecg + slope + ca + thal
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      295      330.03
2      285      203.65 10    126.38 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova(mod2, mod3, test = "Chisq")
Analysis of Deviance Table

Model 1: disease ~ cp + sex + fbs + exang + restecg + slope + ca + thal
Model 2: disease ~ cp + sex + fbs
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      285      203.65
2      293      306.62 -8   -102.97 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova(mod3, mod4, test = "Chisq")
Analysis of Deviance Table

Model 1: disease ~ cp + sex + fbs
Model 2: disease ~ cp + sex + fbs + exang + restecg + slope + ca + thal
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      293      306.62
2      285      203.65  8    102.97 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |
```

Figure 4 Explanation → This figure shows the ANOVA comparisons of the models. The comparisons shown in the figure demonstrate that all the comparisons are very significant

because the p-values are less than 0.001. Each predictor that is added improves the fit of the model. The only predictor that doesn't allow for significant improvement is fasting blood pressure (fbs).

Figure 5: Pearson residual vs Fitted values

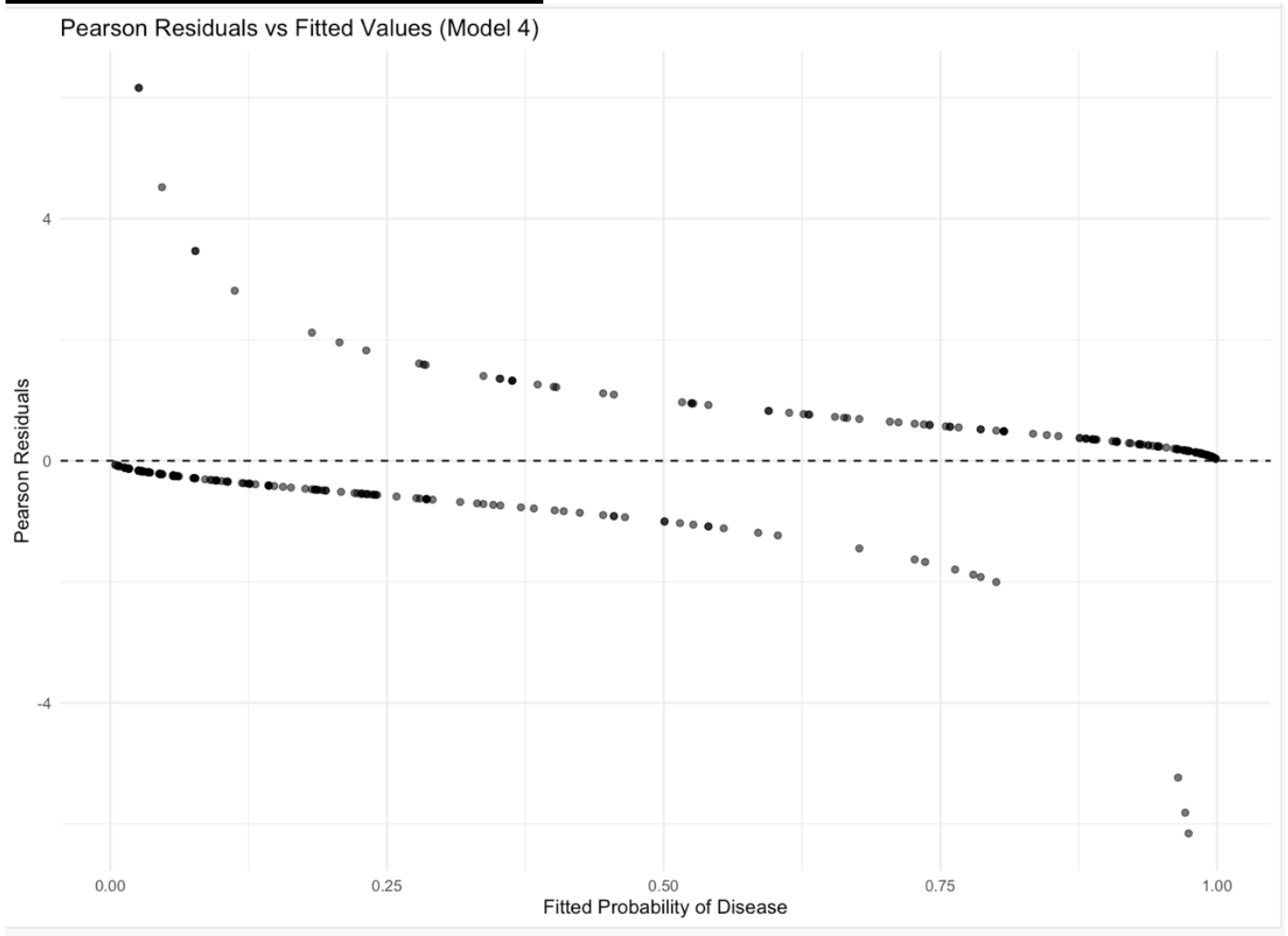


Figure 5 Explanation → This scatterplot shows a Pearson residual of Model 4. Most of the residuals are around 0 which demonstrates a reasonable fit. There are outliers on the top left and bottom right of the scatterplot which demonstrates poorly predicted observations. The slight curve to the plot shows that the model may not fit perfectly but isn't as bad. This acts as a diagnostic that shows that Model 4 is efficient, but not fully perfect in displaying the attributes of the dataset.

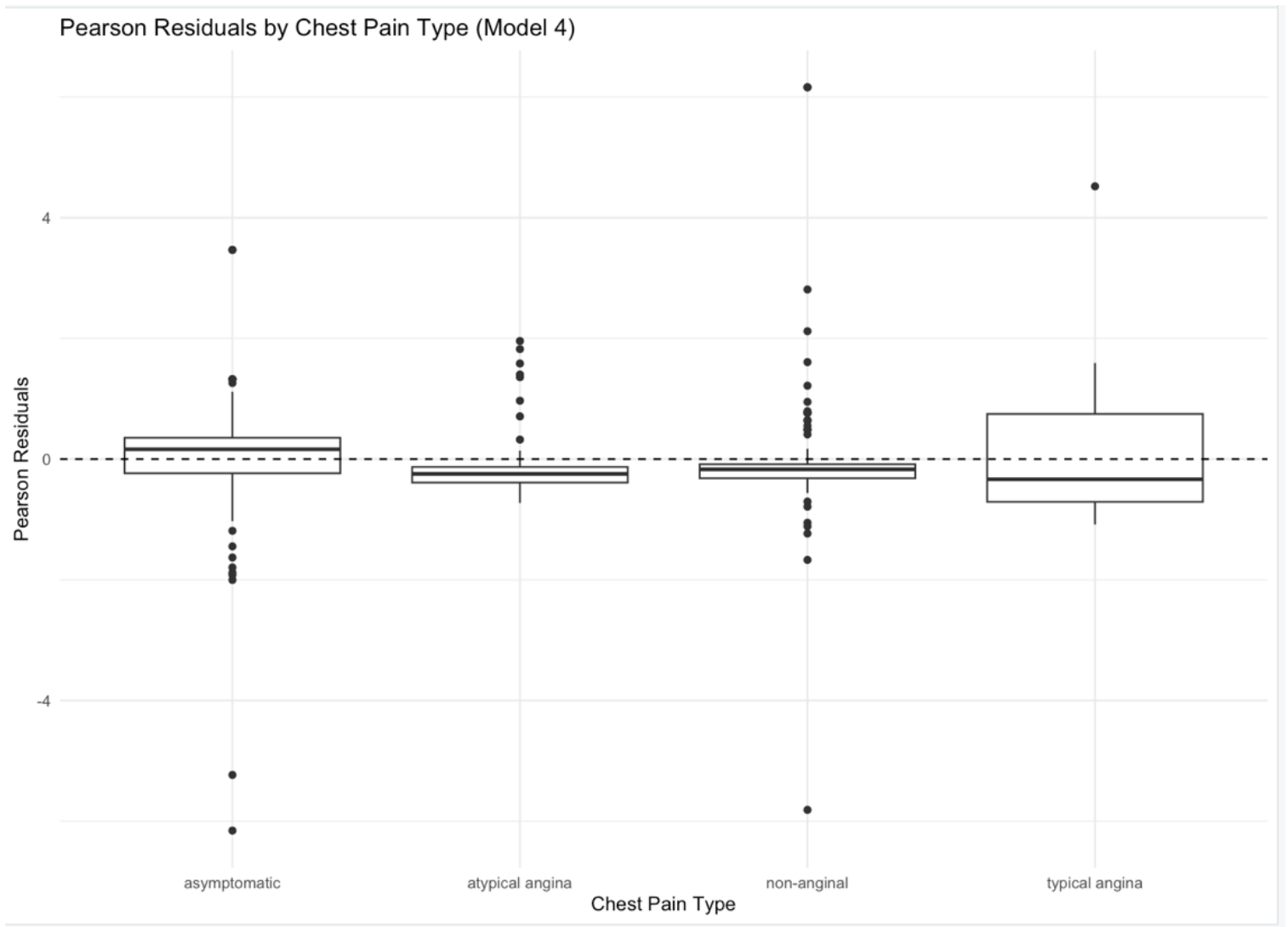
Figure 6: Pearson residuals by CP type

Figure 6 Explanation → This boxplot shows the residual for each type of chest pain. Most of the chest pain groups are focused near 0, which shows that the model is a good and reasonable fit. The spread for typical angina is a bit wider which shows that there is more variability, thus making it harder to predict. There are outliers in all categories. This boxplot differs across each type of chest pain; however, the overall assumptions are met.

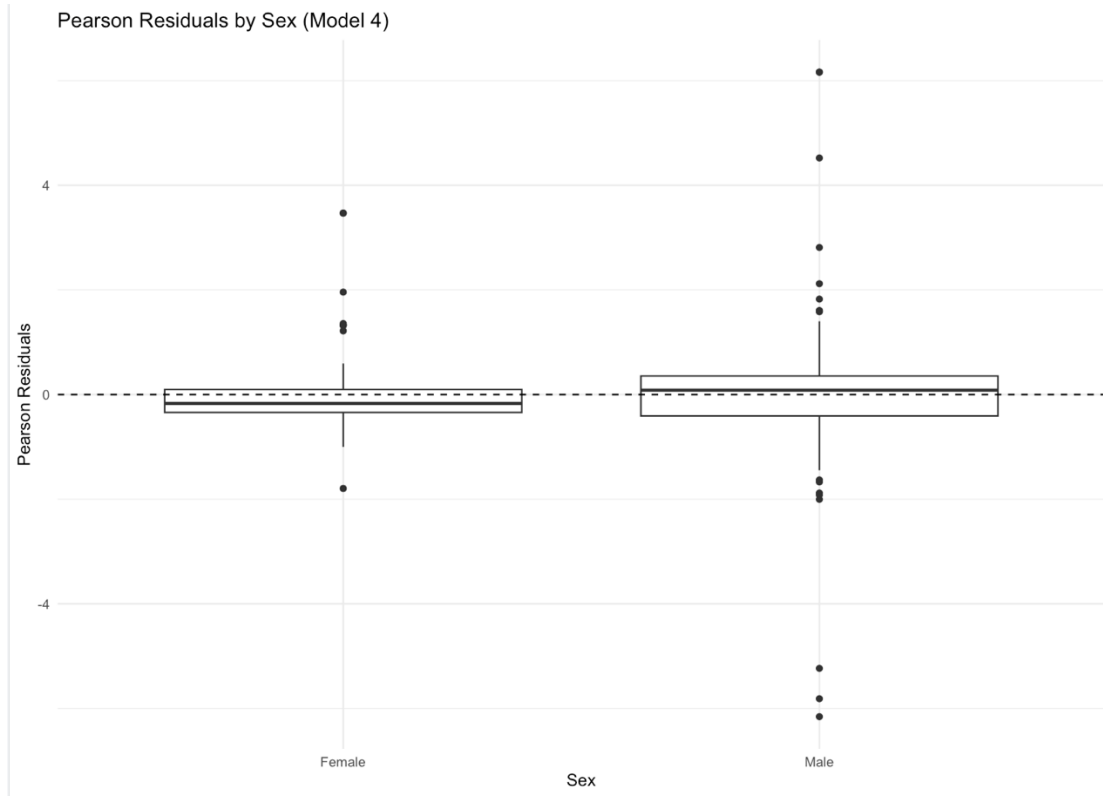
Figure 7: Pearson residual by Sex

Figure 7 Explanation → This boxplot compares the distribution of Pearson residuals for females and males. Both residual groups are close to 0, which shows that the model is a similar fit for both sexes. Males have a little bit of a bigger spread with bigger outliers, in comparison to females. This means that a small number of male observations are harder to predict. Females have more stability in their predictions due to their residual being narrower. This boxplot shows that the larger residuals found in males demonstrates more variability in male patients which can be caused by other external factors.

Figure 8: largest Pearson residual = where Model 4 fits poorly

```
> #4) largest residual (where the model fits the worst)
> heart_diag %>%
+   arrange(desc(abs(pearson_resid_mod4))) %>%
+   select(disease, cp, sex, fbs, exang, restecg, slope, ca, thal,
+         fitted_mod4, pearson_resid_mod4) %>%
+   head(10)
# A tibble: 10 × 11
  disease    cp      sex  fbs  exang restecg  slope    ca  thal  fitted_mod4  pearson_resid_mod4
  <fct>    <fct>    <fct> <fct> <fct> <fct>    <fct> <dbl> <fct>    <dbl>    <dbl>
1 Disease non-anginal Male FALSE FALSE normal  upsloping  0 normal    0.0252      6.16
2 Disease non-anginal Male FALSE FALSE normal  upsloping  0 normal    0.0252      6.16
3 No disease asymptomatic Male FALSE TRUE normal  flat      1 reversible d...  0.974     -6.16
4 No disease non-anginal Male FALSE FALSE normal  flat      3 reversible d...  0.971     -5.81
5 No disease asymptomatic Male TRUE FALSE normal  upsloping  3 reversible d...  0.965     -5.23
6 Disease typical angina Male FALSE FALSE lv hypertrophy upsloping  0 normal    0.0466      4.52
7 Disease asymptomatic Female FALSE FALSE lv hypertrophy upsloping  0 normal    0.0768      3.47
8 Disease asymptomatic Female FALSE FALSE lv hypertrophy upsloping  0 normal    0.0768      3.47
9 Disease non-anginal Male FALSE FALSE normal  flat      0 normal    0.112      2.81
10 Disease non-anginal Male FALSE FALSE lv hypertrophy flat      0 normal    0.182      2.12
```

Figure 8 Explanation → This is a table of the observations with the largest Pearson residual within Model 4. This shows cases where the model has the poorest fit, thus showing how the model's predicted probability is different from the expected outcome. The observations that fit the worst are found in male patients, which validates the previous diagnostic that the model fit varies more in male patients. There are many misfit cases in non-anginal chest pain types because of their low predicted probabilities. The negative residuals connect to asymptomatic patients with high predicted probabilities who don't have heart disease. This means that the model overpredicted the risk in these specific patients. The outliers are consistent with predictors with high risk; however, the actual outcome of the disease goes against the expected outcome of disease. Analyzing cases that are the worst-fitting allows for us to find patterns that the model doesn't necessarily cover. This provides us with more insight on how improvements can be made in the future.

Discussion & Limitations:

The results show that risk for heart disease is best determined by a multivariable model. Using multiple clinical factors to determine the predictive accuracy of the outcome variable proved to be much more efficient than using chest pain type alone. In simple terms, patients who didn't have chest pain, but had abnormal test results, had the highest risk for heart disease. Male sex, abnormal thal results, exercise-induced angina, and high coronary calcium strongly increased the odds for heart disease. The fasting blood sugar predictor added very little to the predictive accuracy of the dataset. Model 4 was the best-fitting model which proved how a multivariable regression model captured the most important and strongest predictors.

Some limitations that are pre-determined is that generalizations can't be made with a bigger overall population because this data has only been taken from one institution. This can lead to selection bias, due to the data not being generalized to all groups. There could also be external factors, like patients having other illnesses or diseases. Since only categorical predictors were used, there is residual confounding from missing certain continuous variables. An overall case analysis could introduce potential bias, and this is seen with the way some diagnostics showed moments where the model didn't have a good fit. Some categories had inconsistencies which included, small sample size, missing values, and formatting discrepancies. This affects the stability of the dataset. An implication is that chest pain characteristics combined with sex and key test results, like thal, ca, and exang, can help identify higher-risk patients. This shows how the multivariable model accounts for all categorical indicators is a lot more fitting for the data analysis. Future work should involve continuous variables and a bigger sample size. I think future work should also have more flexibility in modeling methods. This may include models that show nonlinearity and have interactions. This would show more intricate patterns in the data. Generalized additive models would also show nonlinear effects of predictors without assuming linearity. These new models may be better at addressing certain limitations in my study.

Dataset and Code (code attached separately in R Markdown file):

Link to data source: <https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data?resource=download>